NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

THEORY EXAMINATION

End Semester Examination

Roll No.....

Month and year of examination: **December, 2024**

Subject: **Engineering Practice**

Course Code: **MEIC-102**

Time Allowed: **3 hours**

Programme: **B.Tech.**

Semester: **1st Sem**

Maximum Marks: **50**

- ***Unless stated otherwise, the abbreviations/symbols have their usual meaning in context of the subject.***
- ***Diagrams should be well labeled and drawn properly using the drawing instruments.***
- ***Attempt all the parts of a question at one place maintaining their sequence.***
- ***Be precise and don't provide unnecessary information that has not been asked.***

Q1. Differentiate between DCSP and DCRP in metal arc welding. Which method is preferred for joining thin metal plates and why? **[5+5]**

Q2. What are the different feed mechanisms in lathe machine? Briefly describe any five operations that can be performed using a lathe machine. **[5+5]**

Q3. What are the main goals of legal frameworks in industrial safety, and how do they enhance worker well-being while improving the efficiency of industrial operations? **[10]**

Q4. Draw the schematic diagram of sand mold in casting, depicting different parts of the mold. Explain the importance of runner, riser, sprue and choke in the gating system of a mold. **[4+6]**

Q5. (a) Differentiate between the kiln seasoning and electrical seasoning method. **[4]**

(b) What are the various types of punches used in a fitting shop, and what are their specific applications? **[6]**